CommonsLens

Automatic category suggestions for Wikimedia Commons

TEAM-89 · Problem Statement 3 · Final presentation

Suggest the right categories for a new upload, show the evidence, and say honestly when unsure.

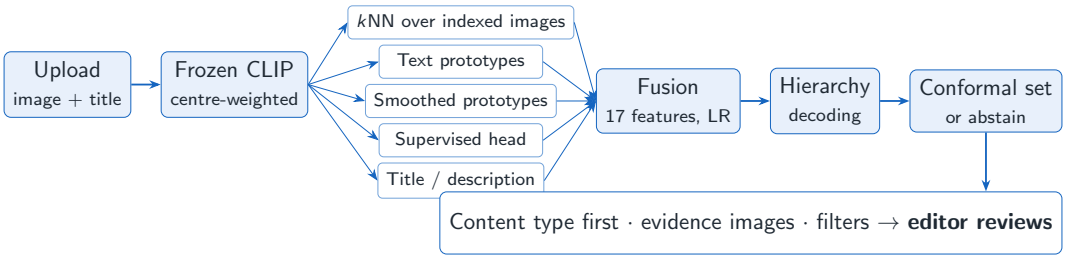
Why this is hard

- **Huge, growing label space:** 24,486 categories in our sample alone; new ones every day
- **Long tail:** 72% of test labels belong to categories with *no* training image
- **Incomplete labels:** a missing category is not a wrong one
- **Editors need trust:** why a category is suggested, and when the system does not know

A fixed classifier with a softmax head fails all four: it needs retraining for every new category, ignores the tail, punishes unlabelled parents, and always answers.

Our answer: categories are *retrievable objects* in a shared image–text space.

Architecture



New categories are suggestible instantly; new images and editor accept/reject update the index without retraining. Fusion runs only on the union of each scorer's top 100 candidates, never on all categories.

What the editor sees

- ⋆ **Content type first:** Object, Living thing, Scene, Event or Abstract concept (or two, e.g. a fruit market)
- **Unsure banner** when the type is below 50%
- ✓ **Suggestion set:** the marked rows contain a correct category with a stated probability; empty set ⇒ "manual categorisation"
- **Why:** the indexed images and signals behind each suggestion
- **Filters:** no maintenance or contest categories; bare country names and over-specific ones (dates, people counts, non-Latin script) hidden unless very sure
- **Report a problem:** testers' reports are stored, de-duplicated and triaged; many fixes came from them
- **Public read-only demo** link for anyone

Data and honest evaluation

- **6,303 Commons files**: 107 topical seeds, 48 region seeds (8 countries per continent), 1,500 uniformly random files
- **Temporal split** by upload time (4,429 / 937 / 937), perceptual-hash and embedding de-duplication
- **Realistic validation:** vocabulary built from training data only, so tuning sees unknown categories like the test set does
  this fixed a validation–test gap of .135 → .014
- **Bias audit** per topic and per region; robustness to blur, JPEG and downscaling; latency and memory
- **77 automated tests**, plus two hand-labelled test packs for the content type

Results on 937 unseen, later uploads

.348
P@1 with a title

.336
R@10 with a title

+53%
fusion vs best single scorer

30 ms
per image on a laptop

F1@5	+meta	image only
Full fusion	**.152**	**.085**
Lexical (title words) only	.099	–
Smoothed prototypes only	.083	–
*k*NN only	.069	–

Recall@5 by category size	labels	R
20+ images	380	.821
5–19	217	.439
1–4	318	.200
unseen	2,412	.003

The tail is the bottleneck: only 36% of test labels are in the vocabulary at all.

What worked, what did not

Smoothed prototypes (M1's core idea)

- Image centroids: prototype recall×10 (.020 → .191)
- Full system unchanged (.152): *k*NN already carries it
- Parent/sibling term hurt the tail (.061 → .030), so it was not used
- Replacing text prototypes broke title matching; kept as a separate signal

Conformal suggestion sets

- Sets adapt: at α=0.4 median 3 categories, beats a fixed top-3 (.518 vs .484)
- 7% of uploads routed to manual categorisation
- Coverage 7–8 points under target: later uploads break exchangeability ⇒ recalibrate on recent uploads

We also caught two evaluation pitfalls: a leaky validation vocabulary, and a display filter inside the ranker that scored every region at zero.

Fairness, robustness, responsibility

F1@5	*n*	+meta	image
Europe	42	.274	.037
Africa	71	.207	.006
S. America	48	.087	.009
Asia	47	.087	.034
N. America	43	.036	.012

Small groups: gaps are indicative.

- **Titles matter for places**: image-only accuracy is near zero in every region
- **Robust**: heavy blur costs 5% with a title, 36% without
- **Licences**: trained only on openly licensed Commons files; ObjectNet, WIDER FACE, DTD and COCO were *not* used (training or commercial restrictions)
- **People**: typed as living things, never identified
- **Human in the loop**: nothing is applied automatically

Content type and the feedback loop

Trained content-type classifier

- 939 openly licensed photos, equal share per source category
- 82.9% on held-out photos, five types
- Living-things pack (250): 91.6% typed Living thing; 88.6% on the 88 hard photos
- Portraits (4 of 14) traced to a collection bug: fixed

Fixed from tester reports

- People pulled to "Marathons" → centre-weighted embedding
- Cathedral typed Object → buildings are Scenes
- Mushroom, portrait → Living thing type
- "3 men 1 woman", dates, Urdu text → hidden unless sure
- "Wiki Loves ..." contest categories → filtered

Scale and next steps

Built to grow

- Frozen encoder: no GPU training; scoring 1 ms per image
- FAISS flat, HNSW or IVF-PQ index by configuration
- New categories and images live without retraining
- MIT licence; runs on a laptop

Next

- Grow the vocabulary and the tail: multilingual names, geotags
- Rolling conformal recalibration on recent uploads
- SigLIP vs CLIP; newest-uploads test set (tools built)
- Human-reviewed score: many “wrong” suggestions are valid

CommonsLens: suggestions editors can check, and honesty when unsure.